

Self-Assembly of Amphiphilic Polymers

A brief description of the project

Amphiphilic molecules consist of a hydrophobic (not favourable with water) and hydrophilic (favourable with water) parts, in solution these molecules can self-assemble into various aggregates, such as the spherical micelle, to protect the hydrophobic section from water. In a spherical micelle, the hydrophobic part is a dense core surrounded by a hydrophilic corona. Micelles naturally form in certain liquids, such as milk, and can be used for the delivery of drug molecules to areas of the body.

These self-assembled aggregates make various contributions to the free energy of a system, for example, because they are deformed away from their natural state there is an energy penalty, so the energy used to deform the aggregates away from their natural state contributes to the free energy of the system. Therefore by minimising the total free energy of the system we are able to predict the shape and other properties of the micelle formed. This is known as scaling theory and we will use this in this project to make predictions about micelle formation. I will also write some code that will plot a graph of fraction copolymer chains aggregated in micelles against the overall copolymer monomers concentration.

Reasons for undertaking the project

I was first introduced to amphiphiles during my condensed matter course in second year and it was immediately something that interested me so when I discovered that I would be able to continue to study them as part of my project it immediately became my first choice. I enjoy working with projects where physics and chemistry intertwine and I enjoy learning about how molecules behave in solution. I also enjoy polymer physics and when we make the predictions about micelle formation we will be making the predictions in mixtures of block copolymers and homopolymers so it is a great opportunity for me to enhance my knowledge on polymer physics. I wanted to do a theoretical project because I enjoy maths and solving equations and doing derivations so that is another reason I chose to do this project. Overall my motivations for choosing this project are that it is something I am extremely interested in and something I believe I will enjoy.

Survey

Title	Date	Author/s	Why I like it
"Theory of critical micelle concentration for solutions of block copolymers"	1983	Ludwik Leibler Henri Orland John C. Wheeler	I like this paper because it identifies a simple mean field model for the formation of micelles and it identifies the critical micelle concentration.

"Micelle Formation and the Hydrophobic Effect"	2004	Lutz Maibaum Aaron R. Dinner David Chandler	I like this paper because it talks about the difference between micelle assembly and macroscopic phase separation. It also describes the hydrophobic effect and the theoretical predictions agree with the experiment.
"Micelle formation in a dilute solution of block copolymers with a polyelectrolyte block complexed with oppositely charged linear chains"	2003	E. Yu. Kramarenko A. R. Khokhlov	This report relates the aggregation number to the concentration of block copolymers. The neutral diblock copolymers are mixed in solution with polyelectrolyte.
"Theory of microphase separation in block copolymer solutions"	1998	M. Olvera de la Cruz	This paper will allow me to learn more about block copolymer solutions and it includes some information on the Flory interaction parameter which I will be using in my project.
"Critical concentration of micelle solutions"	2003	Hans M. Cassel	This will help me when calculating the critical micelle concentration.
"Theory of micelle formation in aqueous solutions"	1974	Charles Tanford	This links micelle size, critical micelle concentration and other properties to size-dependent free energy of micellization with equations. The Hydrophobic component of free energy and its dependence on the size of the micelle is estimated.
"Thermodynamics of Micelle Formation: Prediction of Micelle Size and Size Distribution"	1974 May	Charles Tanford	This paper will help me to learn more about the thermodynamics involved. It also

			portrays a theory for calculating the size of the micelles as well as other properties.
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Action plan

I am going to begin by reading up on previous work done in this field and reading some soft matter books and polymer books. I will spend the first couple of weeks doing this to acquire as much knowledge as possible. I will then spend 2/3 weeks learning about regular solution theory and Boltzmann distribution as these are very important in the project. During the first few weeks I will also be refreshing my knowledge on Latex and practising with it as that is what I will be writing my report in so that it looks better. I will then spend approximately a week on each part of each theoretical equation described in the paper 'Theory of critical micelle concentration for solutions of block copolymers' by Ludwik Leibler, Henri Orland and John C. Wheeler. The last couple of weeks of my project will be slightly computational based, I will be given some code and will input the variables in order to produce a graph fraction copolymer chains aggregated in micelles against the overall copolymer monomers concentration. This will be done for different values of incompatibility degree. My plan is very flexible so if I need to make any changes to it that is not a problem.

Arrangements for weekly meetings with my supervisor

My supervisor have agreed to meet for approximately 20 minutes every Friday to discuss the project and chat about what we have done over the week. We will also make any adjustments to the action plan, if necessary.